

Module 04: Cognition — Behavioral Paradigms for Active Inference

Course: Active Inference 401 | **Unit:** Research Methods | **Audience:** Graduate students / researchers

Learning Objectives

1. Design **behavioral tasks** that dissociate Active Inference computational variables.
2. Analyze **eye-tracking and pupillometry** as measures of prediction and precision.
3. Evaluate paradigms for studying **model-based planning, learning, and decision-making**.

Key Concepts

1. Behavioral Task Design Principles

The core challenge: Active Inference specifies internal computational variables (prediction errors, precision, expected free energy). Behavioral measures (reaction times, choices, eye movements) are indirect reflections of these variables. Good task design makes specific computational variables identifiable from behavior.

Dissociation principle: Design tasks where changes in one computational variable (e.g., precision) produce different behavioral predictions than changes in another variable (e.g., learning rate). If two variables make the same behavioral predictions for a given task, that task cannot dissociate them.

Key task elements:

- **Uncertainty manipulation:** Vary the reliability of cues (high vs. low precision sensory evidence)
- **Volatility manipulation:** Vary how often the environment changes (stable vs. volatile contingencies)

- **Conflict manipulation:** Present situations where prior expectations conflict with sensory evidence
- **Planning depth:** Require multi-step planning with varying horizon lengths

2. Eye-Tracking and Active Visual Inference

Eye movements provide a direct window into Active Inference because saccades ARE active inference — they are actions that sample the environment to resolve uncertainty:

Fixation patterns: Where people look reveals what information they seek. Active Inference predicts fixations toward high-uncertainty, high-information regions (epistemic foraging).

Saccade latency: Faster saccades to locations with higher expected information gain. Slower saccades to predictable regions (no need to check).

Scanpath analysis: The sequence of fixations reveals the brain's hypothesis-testing strategy. Active Inference predicts optimal scanpaths that maximize cumulative information gain.

Visual search: In visual search tasks, Active Inference predicts that search is guided by a generative model of target appearance. Search termination follows when posterior confidence exceeds a precision threshold.

3. Pupillometry

Pupil size is an autonomic marker of several Active Inference computations:

Pupil dilation and surprise: Unexpected events (high prediction error) produce pupil dilation. This indexes the magnitude of free energy changes.

Pupil size and precision: Tonic pupil diameter correlates with (inverse) decision uncertainty. Larger tonic pupils = more uncertain. Connected to noradrenergic locus coeruleus activity, which modulates global precision.

Pupil-linked arousal and volatility: Pupil responses track environmental volatility estimates. Higher estimated volatility → larger tonic pupil.

4. Decision-Making Paradigms

Probabilistic reversal learning: The reward mapping switches periodically. Tests: learning rate adaptation, volatility estimation, perseveration vs. flexibility.

Information sampling tasks: Agents choose between sampling (gathering information) and committing (making a decision). Active Inference predicts sampling continues until expected free energy of committing is lower than expected free energy of continued sampling — a natural stopping rule.

Planning depth tasks: Sequential decision tasks (like chess puzzles or multi-step mazes) where optimal performance requires looking ahead N steps. Active Inference predicts that planning depth correlates with temporal depth of the generative model and available cognitive resources.

Multi-attribute choice: Choosing between options with multiple uncertain attributes. Active Inference predicts attribute-wise information sampling (looking at uncertain attributes first) followed by integration. This can be measured with MouseLab-style paradigms or eye-tracking during choice.

5. Experimental Control and Confounds

Control conditions: Every manipulation must have a matched control that changes the predicted variable without changing others. Example: comparing high-volatility vs. low-volatility blocks controls for difficulty by matching overall reward rates.

Individual differences: Computational phenotyping (Module 02) allows each participant to serve as their own model. Group effects are inferred at the second level (PEB).

Practice and fatigue effects: Randomize condition order, include catch trials, monitor response time distributions for drift.

Summary

Behavioral paradigms for Active Inference include eye-tracking for epistemic foraging, pupillometry for surprise and precision, reversal learning for adaptive belief updating, and

information sampling for planning-as-inference. Good paradigm design dissociates computational variables through targeted experimental manipulations.

Further Reading

- Mirza, M. B. et al. (2018). Scene construction, visual foraging, and active inference. *Frontiers in Computational Neuroscience*, 12, 56.
- Nassar, M. R. et al. (2012). Rational regulation of learning dynamics by pupil-linked arousal systems. *Nature Neuroscience*, 15(7), 1040-1046.
- Schwartenbeck, P. et al. (2015). Evidence for surprise minimization over value maximization in choice behavior. *Scientific Reports*, 5, 16575.